4.4

Costs, Scale of Production & Break-Even

Cambridge IGCSE Business

- Fixed & Variable Costs
- Average & Total Costs
- Economies of Scale
- Diseconomies of Scale
- Break-Even Analysis
- Margin of Safety

What this topic covers

- Classify and calculate fixed, variable, average and total costs
- Understand economies and diseconomies of scale
- Draw, interpret and calculate from break-even charts
- Calculate margin of safety and contribution per unit
- Use cost data to make business decisions

Identifying & Classifying Costs

Fixed · Variable · Total · Average

Costs – Fixed, Variable, Total and Average

Understanding the difference between fixed and variable costs is essential — it affects pricing, profit and business decisions.

Fixed Costs vs Variable Costs

FIXED COSTS

RENT: £5,000/month

Same whether you make
1 or 10,000 units

Also: insurance · salaries · loan repayments

VARIABLE COSTS

RAW MATERIALS

Change with output level
Make 0 units → £0 materials

Also: packaging · energy · piece-rate wages

Key Formulas:

Total Cost:

Fixed Costs + Total Variable Costs

Average Cost:

Total Cost ÷ Number of Units

Total Variable Cost:

Variable Cost per unit × Output

📌 Worked example: Fixed costs £10,000. Variable cost £3/unit. Output: 2,000 units. → Total cost = £10,000 + (£3 × 2,000) = £16,000. Average cost = £16,000 ÷ 2,000 = £8 per unit.

💡 Exam tip: Fixed costs do NOT change with output. Variable costs DO change with output. Average cost falls as output rises — this is economies of scale.

Costs – Using Cost Data to Make Business Decisions

Cost data is not just for accountants — managers use it every day to make better decisions about pricing, production and suppliers.

Setting Price

Must cover all costs and generate profit. Cost-plus pricing adds a % mark-up to total cost per unit.

Bakery: average cost £1.20/loaf → sells at £1.80 (50% mark-up).

Which Product to Make

Compare contribution per unit (selling price – variable cost). Make the product with highest contribution first.

Product A: £5 contribution. Product B: £3. Make A first.

Continue or Stop?

If selling price > variable cost per unit, each unit sold contributes towards fixed costs. Keep producing.

Price £8, VC £5. Contribution = £3. Each sale helps cover £60k fixed costs.

Choosing Suppliers

Compare total costs including delivery, quality and reliability — not just the unit price from each supplier.

Supplier A: £4/unit + £500 delivery. Supplier B: £4.50/unit, free delivery.

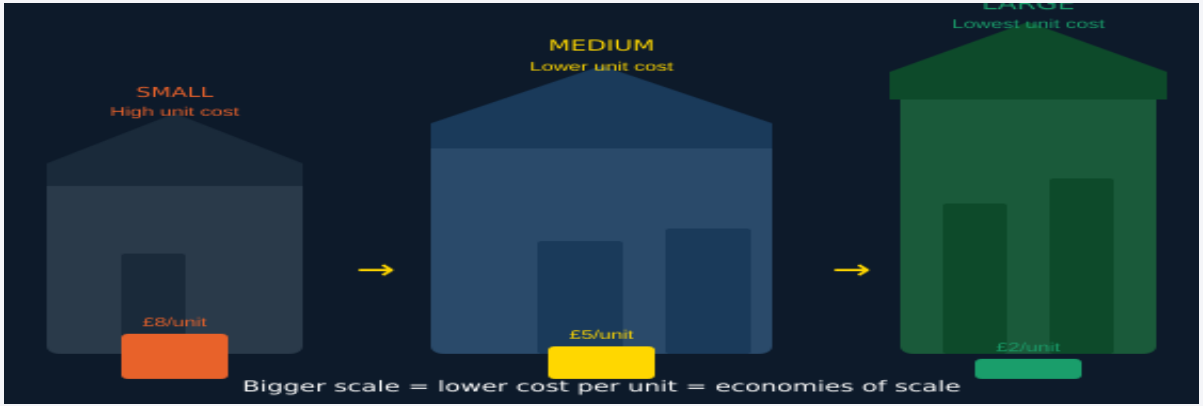
Exam tip: Always use the specific numbers from the case study when making decisions based on cost data. Calculate, then justify your recommendation.

Economies & Diseconomies of Scale

Why bigger isn't always better

Economies of Scale – Definition & Five Types

Economies of scale occur when a business grows and the average cost per unit falls as output increases. This gives large firms a competitive advantage over smaller rivals.



5 types of Economies of Scale:

Purchasing
Buy in bulk → suppliers give discounts · lower cost per unit

Marketing
Spread ad costs over more units · same TV ad for 1m products

Financial
Large firms borrow at lower interest rates · less risk to lenders

Managerial
Specialists hired · 1 HR manager for 10,000 not 10 staff

Technical
Afford expensive machinery · 1 robot serves whole factory

Exam tip: Learn all 5 types with examples. IGCSE often asks: 'Explain one economy of scale this business could benefit from.' Use a specific type, not just 'lower costs'.

Economies of Scale – Real Example: Amazon

Amazon is the world's best example of economies of scale in action. Its massive size gives it cost advantages in every area of the business — advantages no small rival can match.

amazon

Economies of Scale in Action

Purchasing EoS
Buys from suppliers in billions — gets 40% below market price

Technical EoS
750,000 robots reduce fulfilment cost to £2.50 per order

Financial EoS
Borrows at 1.5% — small rivals pay 6%+ on loans

Marketing EoS
Prime Video, ads, Alexa — all share same infrastructure

Result: Amazon can undercut all rivals on price

Diseconomies of Scale

- Poor communication — messages get lost in large organisations
- Lack of employee loyalty — workers feel like a number, not a person
- Weak coordination — departments stop working together effectively
- Lack of management control — too many layers, slow decisions
- Workers become demotivated — feel disconnected from the business

The key point: as a business grows, average costs fall (economies of scale) — but beyond a certain point, average costs can rise again (diseconomies of scale).

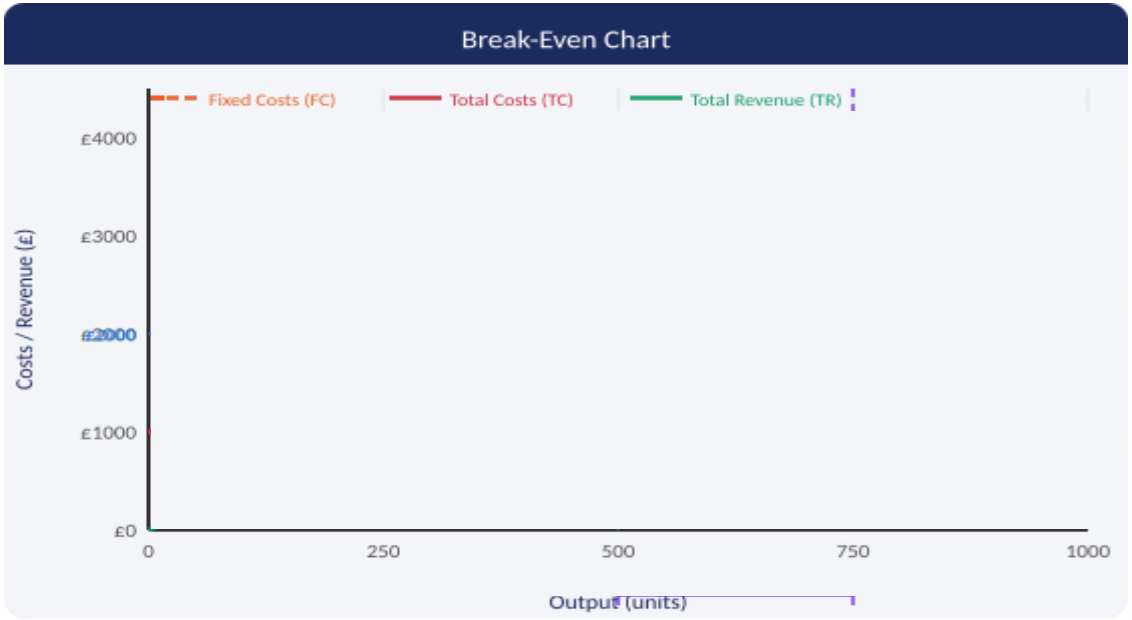
Exam tip: Diseconomies are caused by problems managing a large organisation — communication, motivation and control. Always link to the specific context.

Break-Even Analysis

Charts · Calculation · Margin of Safety · Limitations

Break-Even Analysis – The Chart & Key Concepts

Break-even is the level of output at which total revenue exactly equals total costs — the business makes neither profit nor loss. It is a critical planning tool.



📌 Worked example: Fixed costs £10,000. Variable cost £3/unit. Output: 2,000 units.

Fixed Cost line

Horizontal — same at every output level

Total Cost line

Starts at FC, rises with each unit produced

Revenue line

Starts at 0, rises with each unit sold

Break-even point

Where TC and TR lines cross — zero profit

Margin of safety

Current output minus break-even output

💡 Exam tip: You must be able to draw AND interpret break-even charts. Practice plotting fixed costs, total costs and revenue lines from given data.

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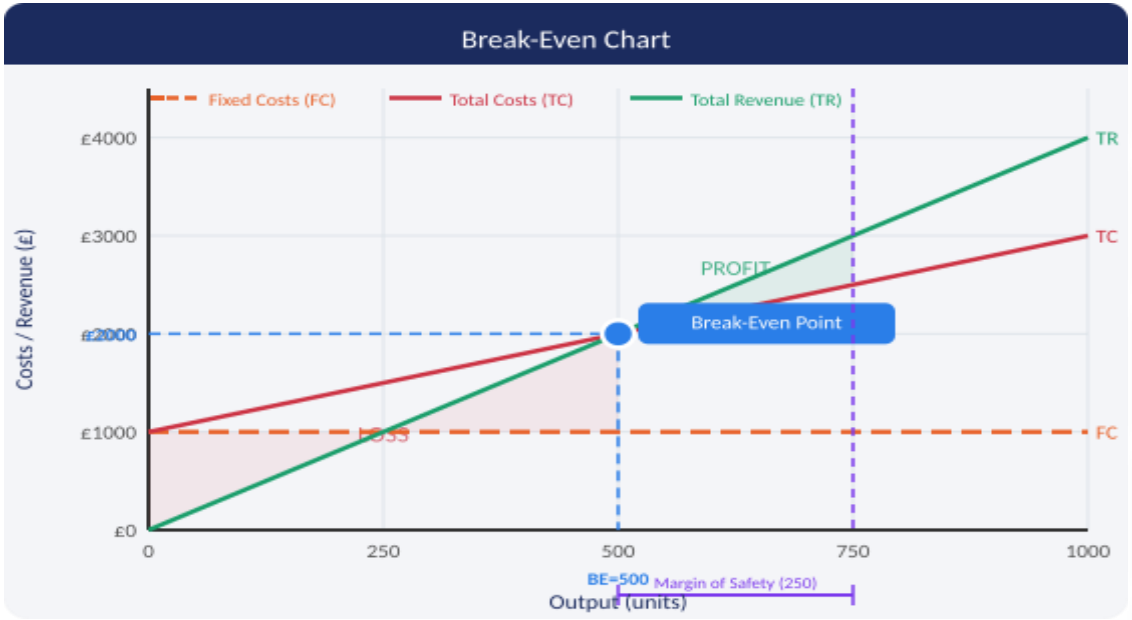

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Break-Even Analysis – Calculations & Worked Example

You must be able to calculate break-even output, contribution per unit and margin of safety. These formulas must be memorised.

Break-Even: Worked Example

Break-Even Output
Fixed Costs ÷ Contribution per unit
= £60,000 ÷ £3 = 20,000 units

Contribution per unit
Selling Price – Variable Cost per unit
= £8 – £5 = £3 per unit

Margin of Safety
Current Output – Break-Even Output
= 30,000 – 20,000 = 10,000 units

Total Revenue at BE
Break-even output × Selling price
= 20,000 × £8 = £160,000

Effect of changes on break-even output:

- ✓ Price rises → TR line steeper → BE point falls (fewer units needed)
- ✗ Price falls → TR line less steep → BE point rises (more units needed)
- ✗ Fixed costs rise → FC and TC lines shift up → BE point rises
- ✓ Variable cost falls → TC line less steep → BE point falls

Limitations of Break-Even

- ✗ Assumes all output is sold — rarely true in practice
- ✗ Assumes costs and prices stay constant — they change
- ✗ Ignores external factors: competition, demand changes
- ✗ Only works for a single product — most firms sell many
- ✗ Based on estimates — data may be inaccurate
- ✗ Doesn't show cash flow timing — profit ≠ cash

Exam tip: Margin of safety = current output – break-even output. A high margin of safety means the business has a comfortable buffer against falling sales.

Key Revision Checklist – 4.4 Costs, Scale and Break-Even

Fixed Costs	Do NOT change with output. Examples: rent, salaries, insurance, loan repayments, insurance
Variable Costs	Change with output. Examples: raw materials, packaging, piece-rate wages, energy per unit
Total Cost	Fixed costs + total variable costs. Average cost = total cost ÷ output (falls as output rises)
EoS Types	Purchasing, marketing, financial, managerial, technical — each reduces average cost per unit as output grows
Diseconomies	Poor communication, lack of loyalty, weak coordination, lack of control — average costs RISE as firm gets too large
Break-Even	Fixed costs ÷ contribution per unit. Contribution = selling price – variable cost per unit. Chart shows TR, TC and FC lines
Margin of Safety	Current output – break-even output. Shows how far sales can fall before a loss is made. Higher = safer.
BE Limitations	Assumes all output sold · costs constant · single product · estimates may be wrong · ignores cash flow timing